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Exam 3

1. Solve each exponential equation

a) $7^x = 2$

$\log 7^x = \log 2$

$x \log 7 = \log 2$

$x = \frac{\log 2}{\log 7} = \boxed{0.35620}$

b) $4 \cdot 7^{4x+1} + 1 = 5$

$4 \cdot 7^{4x+1} = 5 - 1$

$4 \cdot 7^{4x+1} = 4$

$7^{4x+1} = \frac{4}{4}$

$\log 7^{4x+1} = \log\left(\frac{4}{4}\right)$

$(4x+1) \log 7 = \log \frac{4}{4}$

$4x+1 = \frac{\log(4/4)}{\log 7} \approx 0$

$4x = 0 - 1$

$\boxed{x = \frac{-1}{4}}$

c) $3^{4x+1} = 2^{2x+3}$

$\log(3^{4x+1}) = \log(2^{2x+3})$

$(4x+1) \log 3 = (2x+3) \log 2$

$(4x+1) 0.477 = (2x+3) 0.301$

$1.908x + 0.477 = 0.602x + 0.903$

$1.908x - 0.602x = 0.903 - 0.477$

$1.306x = 0.426$

$x = \frac{0.426}{1.306}$

$\boxed{x = 0.326}$

2. The number of bachelor's degrees in the health sciences (in thousands) conferred in the United States can be modeled by an exponential function. In 2012, 163,000 health sciences degrees were conferred, and the number of degrees has risen by 5% each year.

a) Equation

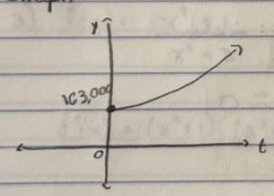
$$y = a(1+r)^t$$

$r = 5\%$ $\qquad$ $a = 163{,}000$

t = the number of years since 2012 $\qquad$ y = number of science degrees conferred

$$y = 163{,}000(1+0.05)^t$$
$$y = 163{,}000(1.05)^t$$

b) Graph



c) Year that the number of degrees first exceed 250,000

$$= 163{,}000(1.05)^t = 250{,}000$$
$$= (1.05)^t = \frac{250{,}000}{163{,}000}$$
$$(1.05)^t \approx 1.533$$
$$\log 1.05^t = \log 1.533$$
$$t \log 1.05 = \log 1.533$$
$$t = \frac{\log 1.533}{\log 1.05} \approx 8.756$$

The year we need to find is $2012 + 8 = 2020$

d) Use the model to estimate how many degrees will be conferred in 2030

In 2030, $t = 2030 - 2012 = 18$

$$y = 103{,}000(1.05)^{18} = 392{,}278$$

3) A company invests in a high-end server cluster for \$65,000. Due to rapid advancements in technology, the business estimates that the equipment's value will decrease by 28% each year.

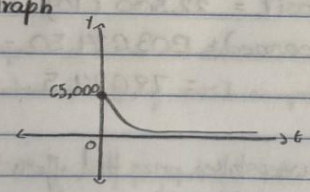
a) Equation

$$y = a(1-r)^t \quad a = 65{,}000 \quad r = 28\%$$

$$y = 65{,}000(1-0.28)^t$$

$$y = 65{,}000(0.72)^t$$

b) Graph



c) Estimated value of the servers after 3 years

$t = 3$

$$y = 65{,}000(0.72)^3 = 24261.12$$

d) After how many years will the equipment be worth less than \$10,000

$$65{,}000(0.72)^t = 10{,}000$$

$$(0.72)^t = \frac{10{,}000}{65{,}000} = 0.153$$

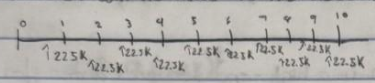
$$\log 0.72^t = \log 0.153$$

$$t(\log 0.72) = \log 0.153$$

$$t = \frac{\log(0.153)}{\log(0.72)} = 5.7147$$

4. A software engineer at a major tech firm decides to maximize their retirement savings. They deposit \$22,500 at the end of each year for 10 years into an investment account that earns 6.5% interest compounded annually.

a) How much will he have in his account after 10 years



$P = 22{,}500$

$r = 6.5\% = 0.065$

$n = 10$

$$FV = P\left[\frac{(1+r)^n - 1}{r}\right] = 22{,}500\left[\frac{1.065^{10} - 1}{0.065}\right]$$

$$= 303{,}624.50$$

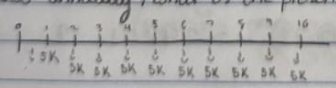
b) How much interest was earned over the 10 years

total deposit $= 22{,}500\,(10) = 225{,}000$

interest earned $= 303{,}624.50 - 225{,}000$

$= 78{,}624.5$

5) You have won a local sweepstakes prize that offers you \$5,000 at the end of each year for 10 years. The sweepstakes company offers you a one-time lump sum payment today instead of the annual payments. If the current interest rate is 5.5% compounded annually, what is the present value of this price?



$P = 5{,}000 \quad r = 5.5\% \quad n = 10$

$$PV = P \cdot \left[\frac{1 - (1+r)^{-n}}{r}\right] \qquad PV = 5{,}000\left[\frac{1 - (1+0.055)^{-10}}{0.055}\right]$$

$$= 37{,}688.129$$

c. A homeowner takes out a $25,000 loan to remodel their kitchen. The bank offers a fixed interest rate of 7.5% compounded monthly for a 48-month term.

a) What can the homeowner expect the monthly payment to be?

$L = 25{,}000 \quad r = 7.5\%/12 = 0.00625 \quad n = 48$

$$M = 25{,}000\left[\frac{0.00625(1+0.00625)^{48}}{(1+0.00625)^{48}-1}\right]$$

$$M = 604.472$$

b) How much will they pay in total for the renovation by the end of the 48 months?

$$604.472(48) = 29{,}014.68$$

7. An entrepeneur takes out a $45,000 loan to launch a boutique coffe shop. The loan has an interest rate of 7.25% compounded monthly.

a) Find the monthly payment and calculate the total interest paid for a 5yr repayment pla

$L = 45{,}000 \quad r = 7.25\%/12 = 0.00604 \quad n = 5(12) = 60$

$$M = 45{,}000\left[\frac{0.00604(1+0.00604)^{60}}{(1+0.00604)^{60}-1}\right] = 896.3286$$

$$\text{Interest} = 896.3286(60) - 45{,}000 = \boxed{8779.716}$$

b) Find the monthly payment and calculate the total interest paid for a 15yr repayment pl

$L = 45{,}000 \quad r = 0.00604 \quad n = 15(12) = 180$

$$M = 45{,}000\left[\frac{0.00604(1+0.00604)^{180}}{(1+0.00604)^{180}-1}\right] = 410.7375$$

$$\text{Interest} = 410.7375(180) - 45{,}000$$
$$= 28{,}932.75$$